README

Compiling the project:

Option 1:

To compile C code with GCC on Windows, install MinGW-w64, which includes the GCC compiler.

1. Download and install **MSYS2** from: msys2.org
2. Open the MSYS2 terminal and run:
   a. pacman -S mingw-w64-ucrt-x86_64-gcc
3. Add GCC to your system PATH: C:\msys64\ucrt64\bin
   a. You need to edit the existing Path variable. A new variable won't help Windows find the command.
   b. Under "System variables", scroll down and find the existing Path variable.
   c. Click on Path -> click "Edit" -> Click "New" and paste: "C:\msys64\ucrt64\bin"
4. Then verify the installation by opening a Command Prompt and running:
   a. gcc –version

To compile the project, use the following command:

```
gcc instance.c optimization.c timer.c utilities.c algorithms.c cw.c vnd.c
main_teacher.c main_lop.c main_vnd.c main.c -o LOP
```

Option 2:

To compile the project, you need to install Visual Studio.

1. Download and install **Visual Studio** from: https://visualstudio.microsoft.com/downloads/
2. Open the LOP.slnx project file in Visual Studio. (You can find it in the root LOP folder.)
3. In the top menu, click Build, then select Rebuild Solution from the dropdown.

Result:

After the build completes, the executable (LOP.exe) will be generated in: ./LOP/x64/Debug/

Running the Project:

1. Run a Single Instance

To run the project, use the following command format:

```
./LOP.exe -m <lop|vnd> -i <input_file> [-b <best_known_value>]
```

Notes:

- m <lop|vnd>: Selects which algorithm to run
  - lop: Runs iterative improvement using all 12 combinations of starting solution (Random, CW), neighbourhood (Transpose, Exchange, Insert), and pivot rule (First-improvement, Best-improvement).
  - vnd: Runs Variable Neighbourhood Descent using the CW starting solution, comparing two neighbourhood orderings (Transpose→Exchange→Insert and Transpose→Insert→Exchange).
- Replace <input_file> with the path to your instance file.
- Replace <value> with the most optimal known cost for that instance.
  - This is optional but allows the program to calculate deviation.
  - The optimal costs can be found in ./LOP/data/best_known/best_known.txt.
- The command must be executed from the directory containing LOP.exe (see above).

Open a command prompt, navigate to the directory containing LOP.exe, and run the command.

Example (Windows - Command Prompt):

```
C:\LOP\x64\Debug> ./LOP.exe -m lop-i ..\..\data\instances\N-be75eec_150 -b
```

2. Run All Instance Files

To run the program for all instance files in the instances folder:

Example (Windows - PowerShell):

```
Get-ChildItem "..\..\data\instances" | ForEach-Object {
  ./LOP.exe -m <lop|vnd> -i $_.FullName
}
```

Linux/Mac:

```
for f in ../../data/instances/*; do ./LOP.exe -m <lop|vnd> -i "$f"; done
```

3. Run All Instances with Best Known Values

To run all instance files in the instances folder with their associated optimal costs:

Example (Windows - PowerShell):

```
$best = @{}
  Get-Content "..\..\data\best_known\best_known.txt" | ForEach-Object {
    $parts = $_ -split "\s+"
    if ($parts.Count -ge 2) { $best[$parts[0]] = $parts[1] }
  }
  Get-ChildItem "..\..\data\instances" -File | ForEach-Object {
    $b = $best[$_.Name]
    if ($b) { ./LOP.exe -m <lop|vnd> -i $_.FullName -b $b }
    else     { ./LOP.exe -m <lop|vnd> -i $_.FullName }
  }
```

Linux/Mac:

```
for f in ../../data/instances/*; do
   name=$(basename "$f")
   b=$(awk -v n="$name" '$1 == n {print $2}' ../../data/best_known/best_known.txt)
   if [ -n "$b" ]; then
      ./LOP.exe -m <lop|vnd> -i "$f" -b "$b"
   else
      ./LOP.exe -m <lop|vnd> -i "$f"
```

Project Structure Overview:

In addition to the executable file located in **./LOP/x64/Debug/**, the project includes two important result directories:

1. implementation_1_results:

This folder contains all files related to **Exercise 1**, which focuses on **basic iterative improvement search algorithms**.

Contents:

- **Instance result files (.csv)**
  For each instance problem provided in ./LOPdata/instances, a CSV file is generated.
  These files follow the naming convention: <instance-name>-result.csv
  For example: N-tiw56r72_250-result.csv
  Each file contains 12 results, corresponding to all combinations of:
  - Pivot rules: first improvement, best improvement
  - Neighborhoods: insert, exchange, transpose
  - Initial configurations: uninformed random picking, Chenery and Watanabe (CW)

- **combine.py**
  A Python script that reads all CSV result files in the current working directory and combines them into a single text file.

- **combined-results.txt**
  The aggregated results produced by running combine.py.

- **analysis_1.txt**
  Contains the output of statistical tests performed in R. This includes the average percentage deviation from the best known solution, the total execution time across all instances per algorithm setup, pairwise Student's t-test p-values (paired, two-sided) comparing all algorithm combinations, and Wilcoxon signed-rank test results identifying pairs with statistically significant performance differences ($p < 0.05$) along with which algorithm performed better in each case.

- **deviation_boxplot_1.pdf**
  A visualization generated in R, showing boxplots of the deviation percentage for each algorithm setup.

2. <u>implementation_2_results:</u>

This folder contains all files related to **Exercise 2**, which focuses on the **Variable Neighborhood Descent (VND)** algorithm.

Contents:

- **Instance result files (.csv)**
  For each instance problem in ./LOPdata/instances, a CSV file is generated.
  These files follow the same naming convention: <instance-name>-result.csv
  Each file contains 2 results, corresponding to the two possible orderings of neighborhood structures:
  - transpose → exchange → insert
  - transpose → insert → exchange

- **combine.py**
  A Python script that reads all CSV result files in the current working directory and combines them into a single text file.

- **combined-results.txt**
  The aggregated results produced by running combine.py.combined-results.txt with the combined results procuced by the python combine.py file

- **analysis_2.txt**
  Contains the output of statistical tests performed in R. This includes the average percentage deviation from the best known solution, the total execution time across all instances per algorithm setup, pairwise Student's t-test p-values (paired, two-sided) comparing all algorithm combinations, and Wilcoxon signed-rank test results identifying pairs with statistically significant performance differences ($p < 0.05$) along with which algorithm performed better in each case.

- **deviation_boxplot_2.pdf**
  A visualization generated in R, showing boxplots of the deviation percentage for each algorithm setup.

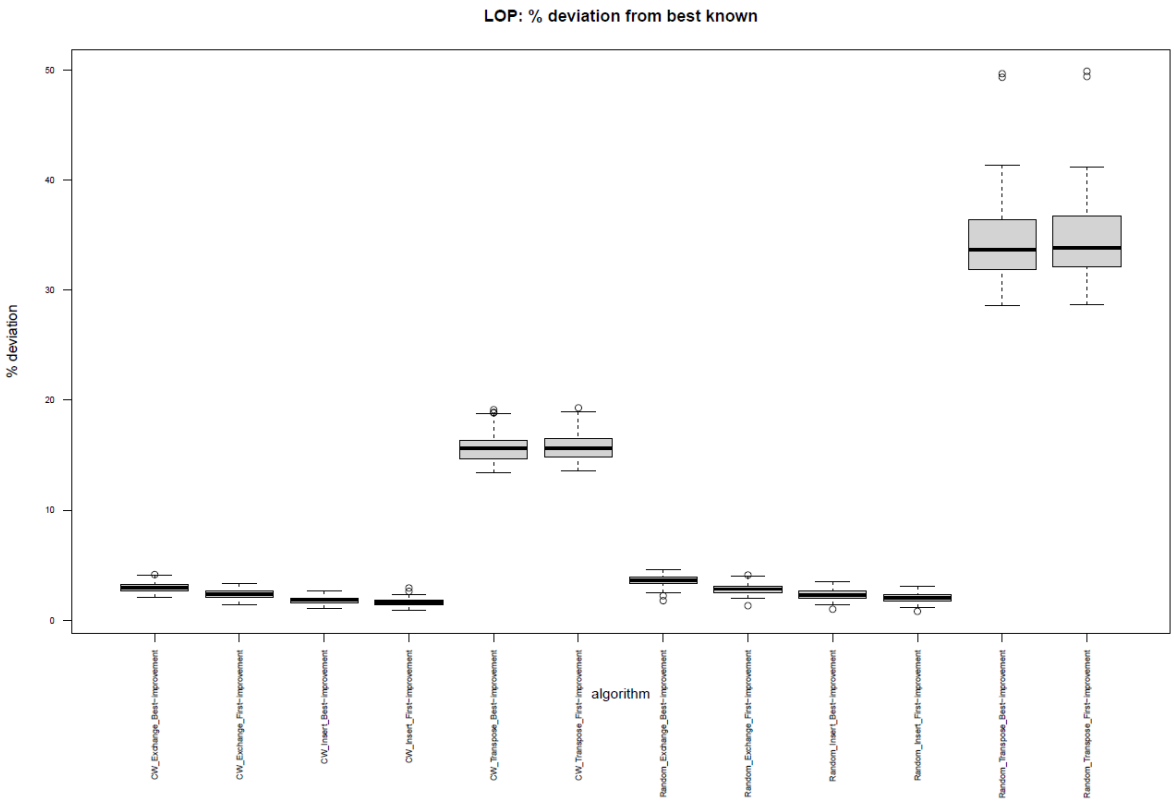
<u>Source Code Location:</u>

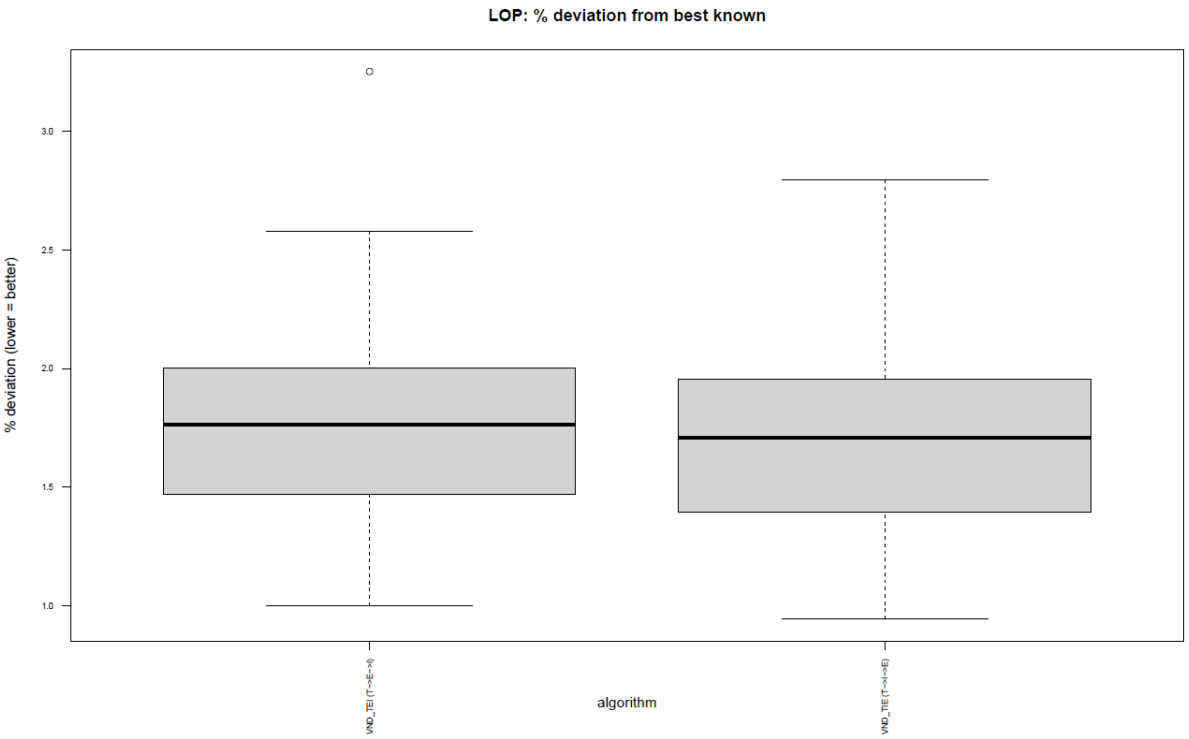
The main source code is organized within a Visual Studio solution (.slnx).
Since the project is named LOP, all source files are located in: **./LOP/LOP/**
This directory contains:

- All **C source files** (.c)
- All **C header files** (.h)
- All **R scripts** used for statistical testing

<u>README:</u>

The README.md file is located in the root of the project directory: **./LOP/**


LOP: % deviation from best known
% deviation
algorithm
CW_Exchange_Best-improvement
CW_Exchange_First-improvement
CW_Insert_Best-improvement
CW_Insert_First-improvement
CW_Transpose_Best-improvement
CW_Transpose_First-improvement
Random_Exchange_Best-improvement
Random_Exchange_First-improvement
Random_Insert_Best-improvement
Random_Insert_First-improvement
Random_Transpose_Best-improvement
Random_Transpose_First-improvement


LOP: % deviation from best known
% deviation (lower = better)
algorithm
VND_TE (T->E->I)
VND_TIE (T->I->E)